# Cognitive Modeling

## Complete Lecture Notes

Week 2: Anatomy of a Model: Signal Detection and Forgetting Curves

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# Chapter 1

# Fundamentals of Cognitive Modeling

## 1.1 Week 2: Anatomy of a Model: Signal Detection and Forgetting Curves

Chapter 1: Fundamentals of cognitive modeling · Lecture 1 h · Lab 2 h This week's running models: equal-variance Gaussian signal detection theory (SDT); exponential and power forgetting curves. Both return in Weeks 3–13.

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### 1.1.1 Key terms

Important terms of this lesson, with their Persian equivalents (repeated even if they appeared in earlier weeks).

| English | فارسی | توضیح کوتاه |
|---|---|---|
| forgetting curve | منحنی فراموشی | کاهش یادآوری با گذشت زمان |
| signal detection theory (SDT) | نظریه‌ی تشخیص علامت | مدلی که حساسیت را از سوگیری پاسخ جدا می‌کند |
| exponential / power function | توانی / تابع نمایی | دو شکل رایج برای منحنی فراموشی و یادگیری |
| parameter | پارامتر | کمیت قابل تنظیم مدل که بین افراد یا شرایط فرق می‌کند |
| deterministic core | هسته‌ی قطعی | بخش بدون تصادف مدل که پیش‌بینی میانگین را می‌دهد |
| noise | نویز | تغییرات تصادفی در شواهد یا در داده |

| English | فارسی | توضیح کوتاه |
|---|---|---|
| linking assumption | فرض پیوند | اینکه کمیت‌های مدل چطور به رفتار قابل مشاهده وصل می‌شوند |
| sensitivity (d′) | حساسیت | فاصله‌ی توزیع علامت و نویز |
| hit | اصابت (hit) | پاسخ «بله» وقتی علامت وجود دارد |
| response bias | سوگیری پاسخ | تمایل به یک پاسخ، مستقل از محرک |
| binomial distribution | توزیع دوجمله‌ای | توزیع تعداد موفقیت در n کوشش |
| noise model | مدل نویز | توزیع داده حول پیش‌بینی مدل |
| simulation | شبیه‌سازی | تولید داده‌ی مصنوعی با اجرای مدل |
| ROC / zROC curve | منحنی ROC / zROC | نرخ hit در برابر نرخ false alarm با تغییر ملاک |
| fit / fitting | برازش | تنظیم پارامترها تا مدل با داده جور شود |
| criterion (c) | ملاک | آستانه‌ی تصمیم؛ نشان‌دهنده‌ی سوگیری پاسخ |
| formal model / formalize | فرمال کردن / مدل فرمال | نوشتن نظریه به شکل معادله یا الگوریتم |
| trial | کوشش (trial) | یک بار انجام تکلیف توسط شرکت‌کننده |
| false alarm | هشدار کاذب (false alarm) | پاسخ «بله» وقتی علامت وجود ندارد |
| estimate / estimator | برآوردگر / برآورد | روش به / مقدار به‌دست‌آمده برای پارامتر دست آوردن آن |
| miss / correct rejection | رد درست / از دست دادن | پاسخ «نه» / پاسخ «نه» با وجود علامت بدون علامت |
| liberal / conservative | محافظه‌کار / آسان‌گیر | «نه» / تمایل به گفتن «بله» |
| free parameter | پارامتر آزاد | پارامتری که از روی داده برآورد می‌شود |
| verbal theory | نظریه‌ی کلامی | نظریه‌ای که فقط با کلمات بیان شده و پیش‌بینی دقیق نمی‌دهد |

### 1.1.2 Learning outcomes

By the end of this week, students will be able to:

1. Decompose a cognitive model into its components: design, free and fixed parameters, deterministic core, noise (data) model, and linking assumptions.
2. Compute $d'$ and $c$ from a 2×2 table of counts, including a correction for hit or false-alarm rates of 0 or 1, and explain why $d'$ separates sensitivity from bias.
3. Implement exponential and power forgetting models with a binomial noise model, and predict qualitatively how each parameter changes the curves.
4. Use simulation and diagnostic plots (ROC/zROC, semi-log and log-log plots) to evaluate whether a model's qualitative signature appears in data.

5. Interpret fitted parameters in psychological terms and identify the limits of the "fit a straight line to transformed data" approach.

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### 1.1.3 Motivation & big picture

Two radiologists. Radiologist R1 flags 93% of scans that really contain a tumor, but also flags 31% of healthy scans. Radiologist R2 flags only 69% of tumors, but just 7% of healthy scans. Their percent correct is identical (81%). Who is the better diagnostician?

Percent correct cannot answer this, because it mixes two different things: how well a person can tell tumors from healthy tissue (*sensitivity*), and how willing they are to say "tumor" (*bias*). Signal detection theory is a small model with two parameters that pulls these apart. As the worked example will show, R1 and R2 have essentially the same sensitivity; they differ only in where they set their criterion. This is the first payoff of modeling: model parameters can measure latent constructs that raw scores conflate.

Ebbinghaus's curve. Since Ebbinghaus's experiments on himself in the 1880s, we know that forgetting is fast at first and then slows down. "Fast, then slow" is a verbal description that both an exponential and a power function satisfy. Yet they imply different things: an exponential means a memory loses a *constant proportion* of its strength per day, at any age; a power function means older memories become *more resistant*. Choosing between them is a question about the nature of memory, and it requires a formal model.

Where is this used?

- Medical diagnosis. Radiologists, pathologists and screening programs are evaluated with SDT and ROC curves, so that a doctor who "sees better" can be told apart from a doctor who simply says "abnormal" more often.
- Memory research. In a recognition test, studied words are "signal" and new words are "noise". $d'$ measures memory strength separately from the tendency to say "old".
- Eyewitness identification. Police line-up procedures are compared with ROC analysis: some procedures do not improve accuracy, they only make witnesses more cautious.
- Perception and hearing tests. Thresholds for seeing or hearing faint stimuli are measured with SDT, so that a cautious listener is not mistaken for a hard-of-hearing one.
- Machine learning. Every binary classifier (spam filter, fraud detector, AI-text detector) has a hit rate and a false-alarm rate, and is evaluated with ROC curves and the area under the curve (AUC). This is the same theory.
- Forgetting curves are used to schedule learning: spaced-repetition apps (e.g. for vocabulary) decide when to show an item again from a model of how fast it is forgotten.

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### 1.1.4 Lecture notes

In plain words This week has one big idea: a model has parts, and each part is a separate assumption. We look at two small models. Signal detection theory explains yes/no decisions under uncertainty. Every trial gives the observer a noisy "feeling of signal". If the feeling is stronger than a personal threshold, the observer says "yes". Two numbers describe the observer: how far apart the "signal" and "no signal" feelings are on average (sensitivity, $d'$), and where the threshold is (criterion, $c$). Two people can have the same percent correct for completely different reasons, and SDT shows which. The forgetting curve describes how recall drops with time. Two formulas, exponential and power, both look like "fast drop, then slow", but they say different things about memory. Choosing between them is the kind of question the rest of the course teaches us to answer with data.

Statistics words in plain language

- Distribution: a description of which values a random quantity takes and how often. A histogram of many measurements approximates it.
- Normal (Gaussian) distribution $\mathcal{N}(\mu, \sigma)$: the bell curve, with mean $\mu$ (center) and standard deviation $\sigma$ (width). About 68% of values fall within $\mu \pm \sigma$ and 95% within $\mu \pm 1.96\sigma$.
- Standard normal: the normal distribution with $\mu = 0$, $\sigma = 1$. A value measured on this scale is called a z-score: "how many SDs above the mean".
- $\Phi(x)$ (CDF): the probability that a standard normal value is *below* $x$. $\Phi(0) = 0.5$, $\Phi(1) \approx 0.84$, $\Phi(1.96) \approx 0.975$. In Python: `norm.cdf(x)`.
- $\Phi^{-1}(p)$ (inverse CDF, "z-transform"): the opposite question: below which value lies a proportion $p$? $\Phi^{-1}(0.84) \approx 1$. In Python: `norm.ppf(p)`.
- Binomial distribution Binomial$(n, p)$: the number of successes in $n$ independent tries, each with success probability $p$ (e.g. words recalled out of 50). Its mean is $np$ and its variance $np(1-p)$.
- Proportion vs. probability: a proportion is what we *observe* (30 of 50 recalled = 0.60); a probability is the model's *underlying* value that the proportion estimates.

Suggested timing (60 min): anatomy of a model 15 · SDT 20 · forgetting curves 15 · simulation as a thinking tool + preview 10.

#### 1. The anatomy of a model

In Week 1 we wrote a model as a data-generating machine, y $\sim p(\text{y} \mid \boldsymbol{\theta}, \mathcal{M})$. In practice, almost every model you will meet in this course can be taken apart into five pieces:

| Component | Question | SDT | Forgetting (exponential) |
|---|---|---|---|
| 1. Design $x_i$ | What is manipulated and measured? | Trial type (signal/noise); response yes/no | Retention interval $t_i$; number of items recalled out of $n$ |

| Component | Question | SDT | Forgetting (exponential) |
|---|---|---|---|
| 2. Parameters $\boldsymbol{\theta}$ | What is free to vary? What is fixed? | Free: $d'$, $c$. Fixed: both SDs = 1, noise mean = 0 | Free: $A$, $\lambda$. Fixed: $n$ |
| 3. Deterministic core $f(x;\boldsymbol{\theta})$ | What does the model predict on average? | $H = \Phi(d'/2 - c)$, $F = \Phi(-d'/2 - c)$ | $f(t) = Ae^{-\lambda t}$ |
| 4. Noise model | How do observations scatter around the prediction? | Hits $\sim$ Binomial$(n_s, H)$; false alarms $\sim$ Binomial$(n_n, F)$ | Recalled $\sim$ Binomial$(n, f(t))$ |
| 5. Linking assumptions | How do model quantities map onto behavior? | "Yes" if internal evidence exceeds the criterion | Recall probability equals memory strength $f(t)$ |

Some points about each component:

- Free vs. fixed parameters. Fixed parameters are not "unimportant". They are *assumptions*. In Week 1 we fixed $t_0 = 300$ ms. In SDT, fixing the noise mean at 0 and both standard deviations at 1 is what makes $d'$ and $c$ estimable at all: shifting or stretching the whole axis would not change any prediction. Such fixes are called scaling constraints.
- Parameter ranges. Every parameter has a meaningful range: $0 \leq A \leq 1$, $\lambda > 0$, $d' \geq 0$ for a non-perverse observer. Respecting these ranges matters when fitting (Week 3).
- The noise model is part of the model. Two researchers with the same $f(t)$ but different noise models (binomial vs. Gaussian) have *different* models and will get different parameter estimates (Week 4). The binomial is natural for counts: its variance $f(1-f)/n$ is largest at $f = 0.5$ and shrinks near 0 and 1.

## 2. Signal detection theory (SDT)

The task. On each trial a stimulus is either *signal* (e.g., a tumor, a studied word, a faint tone) or *noise*. The observer answers "yes" (signal) or "no". The outcomes form a 2×2 table:

| | Respond "yes" | Respond "no" |
|---|---|---|
| Signal trial | Hit | Miss |
| Noise trial | False alarm (FA) | Correct rejection (CR) |

with hit rate $H = \frac{\text{hits}}{\text{hits+misses}}$ and false-alarm rate $F = \frac{\text{FAs}}{\text{FAs+CRs}}$.

The model (intuition). Each trial produces an internal evidence value. It is noisy: even pure noise trials sometimes feel signal-like. Signal trials produce, on average, more evidence. The observer says

"yes" whenever the evidence exceeds a criterion.

The model (formal), equal-variance Gaussian SDT:

$$\text{noise trials: } X \sim \mathcal{N}(0,1), \qquad \text{signal trials: } X \sim \mathcal{N}(d',1), \qquad \text{respond "yes" iff } X > k.$$

- $d'$ is sensitivity: the distance between the two distributions in SD units.
- $k$ is the absolute criterion location. It is more convenient to measure the criterion relative to the midpoint between the distributions: $c = k - d'/2$. With $c = 0$ the observer is unbiased, $c < 0$ is liberal (says "yes" readily), and $c > 0$ is conservative.

Predictions. Using the standard normal CDF $\Phi$:

$$H = P(X > k \mid \text{signal}) = \Phi\left(\tfrac{d'}{2} - c\right), \qquad F = P(X > k \mid \text{noise}) = \Phi\left(-\tfrac{d'}{2} - c\right).$$

Inverting the model. Applying $\Phi^{-1}$ (often written $z$) to both equations and solving gives the familiar formulas:

$$d' = \Phi^{-1}(H) - \Phi^{-1}(F), \qquad c = -\tfrac{1}{2}\left[\Phi^{-1}(H) + \Phi^{-1}(F)\right].$$

Notice what just happened: this is our first parameter estimation. There are two free parameters and two data points ($H$, $F$), so the model can be solved exactly. With more data points than parameters, an exact solution usually does not exist and we need the fitting methods of Weeks 3–4.

The extreme-rate problem. If $H = 1$ or $F = 0$, then $\Phi^{-1}$ is $\pm\infty$ and $d'$ is infinite. This happens often with few trials. A standard fix is the log-linear correction: add 0.5 to every cell count before computing the rates (i.e., add 0.5 to hits and FAs and 1 to the number of trials of each type). We use it by default in the lab.

The ROC curve: SDT's signature. Hold $d'$ fixed and let $c$ vary. The pair $(F, H)$ traces a curve, the receiver operating characteristic (ROC). Every point on one ROC has the same sensitivity but a different bias. Plotting $\Phi^{-1}(H)$ against $\Phi^{-1}(F)$ gives the zROC, which under equal-variance SDT is a straight line:

$$\Phi^{-1}(H) = d' + 1 \cdot \Phi^{-1}(F).$$

So the model makes a *testable* prediction: the zROC is linear with slope 1. ROCs can be measured with confidence ratings (each confidence level acts as a separate criterion). In recognition memory, zROC slopes are typically *below* 1, which motivates the unequal-variance model below.

> Advanced: unequal-variance SDT. If signal trials have $X \sim \mathcal{N}(\mu_s, \sigma_s)$, the zROC is still linear but its slope is $1/\sigma_s$. In that case, $d'$ computed from a single $(H, F)$ pair *changes with the criterion*, so it is no longer a pure sensitivity measure. A common alternative index is $d_a = \mu_s\sqrt{2/(1+\sigma_s^2)}$. Estimating $\sigma_s$ requires more than one criterion, i.e., rating data. You will see this effect directly in lab Part B.

Reading the math The two bell curves. Draw two normal curves on one axis: the "noise" curve centered at 0 and the "signal" curve centered at $d'$. Both have width 1. Draw a vertical line at the criterion $k$. Everything to the right of the line is a "yes".

- $H$ = area of the signal curve to the right of the line. $F$ = area of the noise curve to the right of the line.
- Why $H = \Phi(d'/2 - c)$? The criterion sits at $k = d'/2 + c$ (the midpoint plus $c$). The signal curve is centered at $d'$, so the line is $k - d' = c - d'/2$ SDs from its center. The area to the right of that point is $1 - \Phi(c - d'/2) = \Phi(d'/2 - c)$, because the normal curve is symmetric.
- Numerical example. $d' = 2$, $c = 0$: $H = \Phi(1) = 0.84$ and $F = \Phi(-1) = 0.16$. Now make the observer liberal, $c = -0.5$: $H = \Phi(1.5) = 0.93$, $F = \Phi(-0.5) = 0.31$. Both rates went up, sensitivity is unchanged. This is radiologist R1 of the worked example.

Inverting. $d' = \Phi^{-1}(H) - \Phi^{-1}(F)$: convert each rate into a z-score and subtract. For R1: $1.476 - (-0.496) = 1.97$. $c$ is minus the average of the two z-scores: if both rates are high (both z-scores positive), $c$ is negative, i.e. liberal.

The zROC line. $\Phi^{-1}(H) = d' + \Phi^{-1}(F)$ is just the previous equation rearranged. In z-units, moving the criterion moves both z-scores by the same amount, so the points lie on a line with slope 1 and intercept $d'$.

For an engineer SDT *is* the detection theory of radar and communications, imported into psychology in the 1950s.

| Psychology | Telecom / radar |
|---|---|
| noise trial / signal trial | hypothesis $H_0$ (noise only) / $H_1$ (signal + noise) |
| internal evidence $X$ | test statistic, e.g. matched-filter output |
| criterion $k$ (or $c$) | detection threshold |
| hit rate $H$ | probability of detection $P_D$ |
| false-alarm rate $F$ | probability of false alarm $P_{FA}$ |
| sensitivity $d'$ | deflection $(\mu_1 - \mu_0)/\sigma$; $d'^2$ plays the role of an SNR |
| ROC curve | receiver operating characteristic (same name, same plot) |
| liberal / conservative observer | low / high threshold (high / low $P_{FA}$) |

Differences in emphasis: an engineer usually *chooses* the threshold (e.g. Neyman–Pearson: fix $P_{FA}$, maximize $P_D$). A psychologist *observes* a person's $P_D$ and $P_{FA}$ and *infers* both the SNR ($d'$) and the threshold ($c$) the person used. The criterion $c$ is measured from the midpoint between the two means, so $c = 0$ means "no bias". The optimal threshold depends on prior probabilities and costs exactly as in Bayesian detection, and people often move their criterion in the right direction when payoffs change (Week 4).

### 3. Forgetting curves

Data. At each retention interval $t_i$ (e.g., in days), the participant is tested on $n$ items and recalls $y_i$ of them.

Two candidate deterministic cores:

| | Exponential | Power |
|---|---|---|
| Equation | $f(t) = A\,e^{-\lambda t}$ | $f(t) = A\,(1+t)^{-d}$ |
| Parameters | $A$: initial level; $\lambda$: decay rate | $A$: initial level; $d$: decay exponent |
| Linearizing plot | $\ln f$ vs. $t$ is a straight line with slope $-\lambda$ | $\ln f$ vs. $\ln(1+t)$ is a straight line with slope $-d$ |
| Relative loss rate $-f'(t)/f(t)$ | $\lambda$ (constant) | $d/(1+t)$ (decreasing with age) |
| Psychological reading | A memory loses the same *proportion* of its strength per unit time, whatever its age | Older memories lose strength more slowly; they become more resistant |

The relative loss rate (in survival analysis, the *hazard*) is the key interpretive difference. It is why the choice between the two functions is a theoretical question, not a curve-fitting detail.

Noise model. Recall counts are binomial:

$$y_i \sim \text{Binomial}\big(n, f(t_i;\boldsymbol{\theta})\big).$$

With $n = 50$ items and $f = 0.5$, the SD of the observed proportion is $\sqrt{0.5 \cdot 0.5/50} \approx 0.07$. That is large compared with the differences between models over short intervals, which is a first hint that design (which intervals, how many items) determines whether data can discriminate between models.

> Advanced: the "+1" in the power model is not innocent. $A\,t^{-d}$ is infinite at $t = 0$, so a shift is added. But $(1 + t)$ depends on the unit of $t$: measuring in hours instead of days changes the shape of the curve, not just the scale of the axis. Some authors therefore write $A\,(1 + bt)^{-d}$ with a free scale parameter $b$, which adds a third parameter and raises the model comparison questions of Weeks 6–7. Keep the unit fixed (days) throughout this course.

Reading the math

- $f(t) = Ae^{-\lambda t}$: $A$ is the recall level right after study ($t = 0$), and $\lambda$ says how fast it falls. With $\lambda = 0.1$ per day, recall is multiplied by $e^{-0.1} \approx 0.905$ every day: about 9.5% of what is left is lost each day, on day 1 and on day 100 alike.
- $f(t) = A(1+t)^{-d}$: going from day 0 to day 1 multiplies recall by $2^{-d}$; going from day 9 to day 10 multiplies it only by $(10/11)^d$, close to 1. Old memories hardly change any more.

- Relative loss rate $-f'(t)/f(t)$: "what fraction of the remaining memory is lost per day *right now*". Constant ($\lambda$) for the exponential, shrinking ($d/(1+t)$) for the power function. This single number is the psychological difference between the models.
- $y_i \sim \text{Binomial}(n, f(t_i))$: at delay $t_i$ the participant recalls $y_i$ of $n$ items, and each item is recalled with probability $f(t_i)$. The observed proportion $y_i/n$ scatters around $f(t_i)$ with SD $\sqrt{f(1-f)/n}$, about 0.07 for $n = 50$.

For an engineer

- The exponential curve is the step response of a first-order system (an RC circuit discharging): one time constant $\tau = 1/\lambda$. A constant relative loss rate is the "memoryless" property you know from exponential lifetimes in reliability theory.
- The power curve has no single time constant. A useful picture: a sum of many exponentials with *different* time constants (fast and slow components) looks like a power law. Week 5 shows this exactly, and it is one candidate explanation of power-law forgetting.
- On a log–log plot a power law is a straight line; on a semi-log plot an exponential is a straight line. Engineers use the same trick to identify system orders and $1/f$ noise.
- The binomial noise model is the counting statistics of a detector: like counting bit errors out of $n$ bits, the uncertainty of a proportion shrinks with $1/\sqrt{n}$.

### 4. Simulation as a thinking tool

Before fitting anything, simulate. Simulation answers questions such as:

- What can the model produce, and what can it *not* produce? Equal-variance SDT with $d' > 0$ can never produce $H < F$, and it can never produce a zROC slope different from 1. Both exponential and power curves are always decreasing, so they cannot capture *reminiscence* (recall improving over time). The set of things a model cannot do is what makes it testable.
- Which parameters matter where? $A$ moves the whole forgetting curve; $\lambda$ or $d$ controls the shape.
- Can the design discriminate between models? If two models make nearly identical predictions at the intervals we plan to test, no fitting method can tell them apart (lab Exercise 4; Assignment A1).

### 5. Preview: estimation by linearization, and why it is not enough

A traditional way to fit forgetting curves is to log-transform the observed proportions and fit a straight line by ordinary regression. It is fast and intuitive, and we try it in the lab. It has three problems that motivate Weeks 3–4:

1. $\ln(0)$ is undefined, so an interval with zero recalls breaks the method.
2. The log transform distorts the noise: the scatter of $\ln \hat{p}$ is much larger for small proportions, but regression treats all points as equally reliable.

3. The line is fit on the log scale, but we care about errors on the original (proportion) scale.

The principled alternative is to fit the model directly to the untransformed data, by minimizing squared error (Week 3) or maximizing the binomial likelihood (Week 4).

For the instructor

- Open with the radiologist puzzle (Motivation) and let students vote on who is better before showing any math. Most will pick by percent correct, then realize it is identical.
- Draw, don't derive. The two overlapping bell curves with a movable vertical line explain $H$, $F$, $d'$ and $c$ better than any formula. Move the line left and right on the board and ask what happens to $H$ and $F$; then pull the curves apart and ask again.
- Use $\Phi$ tables or Python live. Compute R1's $d'$ with `norm.ppf` in front of the class. Students who are new to statistics need to see that $\Phi^{-1}$ is "just a lookup".
- Connect to something they know: a COVID or pregnancy test has a hit rate (sensitivity) and a false-alarm rate (1 − specificity). Medical "sensitivity" is the hit rate, *not* $d'$: point out this naming clash.
- Typical questions.
  - *"Why assume normal distributions?"* It is a convenient and often adequate assumption; the zROC test (slope 1) checks it, and the unequal-variance model relaxes it.
  - *"Is a liberal criterion bad?"* No: the best criterion depends on costs and base rates.
  - *"Exponential or power, which is right?"* For many datasets the power function fits better, but averaging over people can create a power shape (Week 5). Keep the question open.

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### 1.1.5 Worked example

#### Part 1: Two radiologists (and a third)

Each radiologist reads 100 tumor scans and 100 healthy scans.

| Observer | $H$ | $F$ | $\Phi^{-1}(H)$ | $\Phi^{-1}(F)$ | $d'$ | $c$ | Percent correct |
|---|---|---|---|---|---|---|---|
| R0 (neutral) | 0.84 | 0.16 | 0.994 | −0.994 | 1.99 | 0.00 | 84% |
| R1 (liberal) | 0.93 | 0.31 | 1.476 | −0.496 | 1.97 | −0.49 | 81% |
| R2 (conservative) | 0.69 | 0.07 | 0.496 | −1.476 | 1.97 | +0.49 | 81% |

Step by step for R1: 1. $\Phi^{-1}(0.93) = 1.476$ and $\Phi^{-1}(0.31) = -0.496$ (`scipy.stats.norm.ppf`). 2. $d' = 1.476 - (-0.496) = 1.972$. 3. $c = -\frac{1}{2}(1.476 + (-0.496)) = -0.490$, which is negative, so R1 is liberal. 4. Check by plugging back in: $H = \Phi(1.972/2 + 0.490) = \Phi(1.476) = 0.93$. ✓

Interpretation. All three radiologists have essentially the same sensitivity ($d' \approx 2$). R1 and R2 have the *same* percent correct but *opposite* biases. Which bias is better is not a question of perception but of costs: missing a tumor is usually far worse than a false alarm, so a liberal criterion may be the *desirable* one. SDT lets us ask "can they see it?" and "how willing are they to say so?" separately.

The extreme-rate fix. A participant scores 50/50 hits and 3/50 false alarms. Then $H = 1$ and $d' = \infty$. With the log-linear correction: $H = 50.5/51 = 0.990$, $F = 3.5/51 = 0.069$, so $d' = 2.33 - (-1.49) = 3.82$.

### Part 2: Exponential vs. power forgetting

Take $A = 0.95$ for both models, $\lambda = 0.25$/day for the exponential and $d = 0.6$ for the power model.

| $t$ (days) | 0 | 1 | 2 | 7 | 14 | 30 |
|---|---|---|---|---|---|---|
| Exponential $0.95\,e^{-0.25t}$ | 0.950 | 0.740 | 0.576 | 0.165 | 0.029 | 0.001 |
| Power $0.95\,(1+t)^{-0.6}$ | 0.950 | 0.627 | 0.491 | 0.273 | 0.187 | 0.121 |

Half-life (time until recall halves from its initial level): - Exponential: $e^{-\lambda t} = 1/2 \Rightarrow t_{1/2} = \ln 2/\lambda = 0.693/0.25 = 2.77$ days. - Power: $(1+t)^{-d} = 1/2 \Rightarrow t_{1/2} = 2^{1/d} - 1 = 2^{1.667} - 1 = 2.17$ days.

Interpretation. Over the first two days the curves are fairly similar: they differ by about 0.1, which is only about 1.5 binomial SDs with $n = 50$ items. After a month they disagree completely: the exponential model says practically everything is forgotten (0.1%), while the power model says 12% is still retained. Long retention intervals are where these models can be told apart, a design lesson you will use in Assignment A1.

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### 1.1.6 Lab (2 hours)

#### Goals

1. Implement SDT as a generative model, simulate yes/no experiments, and estimate $d'$ and $c$.
2. Use ROC and zROC plots to evaluate the equal-variance assumption.
3. Implement exponential and power forgetting with binomial noise, and explore their parameters.
4. Fit both forgetting models by linearization, evaluate them with residuals, and interpret the result.

Timing: Part A 30 min · Part B 25 min · Part C 30 min · Part D 25 min · remaining time: exercises.

#### Setup

Code · کد: `code/week02.py`, cell 1 · سلول 1

#### Part A: SDT as a generative model (30 min)

Code · کد: `code/week02.py`, cell 2 · سلول 2

Simulate one participant and recover the parameters:

Code · کد: `code/week02.py`, cell 3 · سلول 3

Expected output:

```
Counts: {'hits': 68, 'misses': 32, 'fas': 11, 'crs': 89}
Predicted H = 0.674, F = 0.147
Estimated d' = 1.67 (true 1.5),  c = 0.37 (true 0.3)
```

One simulated participant tells us little about how good the *estimator* is. Let's repeat the experiment many times: this is a preview of the parameter recovery logic of Week 7.

Code · کد: `code/week02.py`, cell 4 · سلول 4

Expected output:

```
            d_hat         c_hat
             mean    std   mean    std
n_per_type
20          1.500  0.449  0.296  0.224
50          1.489  0.290  0.295  0.143
200         1.502  0.142  0.299  0.071
```

What to notice: the estimates scatter around the true values, and the scatter shrinks as the number of trials grows. With 20 trials per type, the SD of $\hat{d}'$ is about 0.45, so individual estimates are often off by $\pm 0.5$ or more, and roughly $4\times$ more trials are needed to halve the scatter. On average the estimates are on target here. With higher $d'$ and few trials, however, extreme rates become common, and the correction then pulls estimates toward 0 (try `true_d=3` with 20 trials). Keep this in mind whenever you compare $d'$ across groups tested with different numbers of trials.

### Part B: ROC curves and the equal-variance assumption (25 min)

First, the model's predicted ROCs:

Code · کد: `code/week02.py`, cell 5 · سلول 5

Now simulate a confidence-rating experiment. Responses are given on a 6-point scale (1 = "sure noise" … 6 = "sure signal"), which amounts to five criteria at once. We generate data from an equal-variance observer and from an unequal-variance observer whose signal distribution has SD 1.5.

Code · کد: `code/week02.py`, cell 6 · سلول 6

Expected output:

```
sigma_signal = 1.0: zROC slope = 0.96 (predicted 1.00); d' at each
↪  criterion = [1.52 1.48 1.48 1.43 1.43]
sigma_signal = 1.5: zROC slope = 0.67 (predicted 0.67); d' at each
↪  criterion = [1.46 1.34 1.16 0.96 0.84]
```

Evaluate and interpret. For the equal-variance observer, the zROC slope is close to 1 and $d'$ is about the same at every criterion: the model's assumption holds. For the unequal-variance observer, the slope is close to $1/1.5 = 0.67$ and $d'$ changes systematically with the criterion. If we had collected only yes/no data from such an observer, a group difference in "$d'$" could be produced purely by a difference in bias. Checking a model's signature prediction (here: zROC slope = 1) is a basic form of model evaluation.

### Part C: Forgetting curves with binomial noise (30 min)

> Code · کد: `code/week02.py`, cell 7 · سلول 7

Explore the parameters: how do $A$ and the decay parameter change each curve?

> Code · کد: `code/week02.py`, cell 8 · سلول 8

Now generate a simulated experiment. The data come from the *power* model, with 50 items tested at each of 8 retention intervals:

> Code · کد: `code/week02.py`, cell 9 · سلول 9

Expected output:

```
       t  recalled   n  p_obs
0   0.25        42  50   0.84
1   1.00        30  50   0.60
2   2.00        26  50   0.52
3   4.00        25  50   0.50
4   7.00        13  50   0.26
5  14.00        13  50   0.26
6  21.00        13  50   0.26
7  30.00        14  50   0.28
```

### Part D: Fit by linearization, evaluate, interpret (25 min)

Fit. Fit each model by a straight-line regression on the log scale (Lecture §5).

Code · کد: `code/week02.py`, cell 10 · سلول 10

Evaluate. Look at the fitted curves *and* the residuals. A good model leaves residuals that scatter randomly around zero; a systematic pattern means the model's *shape* is wrong.

Code · کد: `code/week02.py`, cell 11 · سلول 11

Expected output:

```
exp: A = 0.550, rate = 0.032, RMSD = 0.1308
pow: A = 0.797, rate = 0.370, RMSD = 0.0628
true (power): A = 0.9, d = 0.5
```

Interpret (write 2–3 sentences for each in your notebook):

1. Which model describes the data better, and what does the residual pattern of the *other* model look like? (Look for an inverted-U or U shape: the exponential overshoots at intermediate delays and undershoots at the long ones, or the reverse.)
2. Using the fitted parameters, what is each model's relative loss rate after 1 day and after 20 days? What does this mean for memory?
3. The power model is the true model here, yet its estimates ($\hat{A} \approx 0.80$, $\hat{d} \approx 0.37$) are noticeably off the true values ($A^* = 0.9$, $d^* = 0.5$). Name two reasons. (Hints: look at the last three data points; think about what the log transform does to the weights of small proportions.) In Week 3 we refit these same data on the original scale and compare.

### Exercises (increasing difficulty)

Exercise 1 $\star$ (extreme rates). A participant has 40/40 hits and 0/40 false alarms. Compute $d'$ without and with the log-linear correction. What does the corrected $d'$ represent, and why should you report the correction you used?

Exercise 2 $\star$ (half-lives). Using your fitted parameters from Part D, compute each model's half-life. Then compute how long it takes recall to drop to 10% of its initial level. Where do the models disagree most?

Exercise 3 $\star\star$ (Ebbinghaus's data). The classic savings data from Ebbinghaus's 1885 self-experiments are often reported as follows [verify against the original source before using them in class]:

| Interval | 20 min | 1 h | 8.8 h | 1 day | 2 days | 6 days | 31 days |
|---|---|---|---|---|---|---|---|
| Savings (%) | 58.2 | 44.2 | 35.8 | 33.7 | 27.8 | 25.4 | 21.1 |

Convert the intervals to days, fit both models by linearization, and compare their residuals. Which shape fits better? Then redo the analysis with time measured in hours. Does the power fit change? Explain, using the Advanced box in Lecture §3.

Exercise 4 ⋆⋆ (which design can tell the models apart?). Choose $(A, \lambda)$ and $(A, d)$ so that the two models predict the *same* recall at $t = 1$ and $t = 7$ days. (Hint: two equations in two unknowns per model; take logs.) Then plot both curves from $t = 0$ to 60 days and mark where their difference exceeds 2 binomial SDs with $n = 50$. Which retention intervals would you test?

Exercise 5 ⋆⋆⋆ (Advanced: zero counts). Generate data from the exponential model with $A = 0.9$, $\lambda = 0.3$ at the intervals in Part C, with $n = 20$ items. How often (out of 1000 simulated datasets) does at least one interval produce zero recalls, which breaks `fit_linearized`? Propose and implement a quick fix (e.g., adding 0.5 to counts) and check whether it biases the estimate of $\lambda$. Explain why this is a sign that we need a method that works on the original scale (Weeks 3–4).

#### Lab check-off

Show the instructor or TA: (1) your SDT recovery table from Part A, (2) the two empirical zROCs with their slopes from Part B, and (3) the residual plot from Part D with a one-sentence interpretation.

---

### 1.1.7 Connection to AI & LLMs

AI-text detectors are observers in an SDT task. Universities increasingly use tools that score a text for "how likely it was written by AI". Each text is either AI-written ("signal") or human-written ("noise"), and the tool says "AI" when its score exceeds a threshold. This is exactly the SDT setting of this week: the tool has a sensitivity $d'$ (how well the score separates the two kinds of texts) and a criterion $c$ (the threshold the user chooses). A false alarm means a student is wrongly accused, so the choice of criterion is an ethical decision, not only a technical one.

Code · كد: `code/week02.py`, cell 12 · سلول 12

Expected output:

```
threshold 0.0: H = 0.84, F = 0.49, d' = 1.03, c = -0.50
threshold 0.6: H = 0.70, F = 0.28, d' = 1.11, c = 0.03
threshold 1.2: H = 0.51, F = 0.09, d' = 1.35, c = 0.66
```

Moving the threshold changes $c$ a lot (from $-0.50$ to 0.66), while $d'$ stays near its true value of 1.2. The variation in $d'$ is sampling noise from 200 texts per group.

Try it (no API needed). - ⋆ Suppose 10% of submitted essays are AI-written. At threshold 1.2, what is $P(\text{AI-written} \mid \text{flagged})$? (Use Bayes' rule with $H$ and $F$.) Would you act on a flag? - ⋆⋆ Plot the ROC of the detector (all thresholds from $-2$ to 3) and its zROC. Is the slope close to 1? Repeat with

human-text scores of SD 1.5 (an unequal-variance detector). - ⋆⋆ Discussion: detectors have been reported to flag texts by non-native English writers more often [verify: Liang et al., 2023, *Patterns*]. In SDT terms, what would that mean, and what should a university do?

---

### 1.1.8 Common mistakes & misconceptions

1. "Percent correct is good enough." It confounds sensitivity and bias (see the radiologists). Use $d'$ and $c$ whenever responses can be biased.
2. Computing $d'$ as $\Phi^{-1}(H - F)$ or from percentages. The formula is $\Phi^{-1}(H) - \Phi^{-1}(F)$, with rates as proportions between 0 and 1.
3. Forgetting the extreme-rate problem. $H = 1$ or $F = 0$ gives an infinite $d'$; Python returns `inf` without an error. Always use (and report) a correction.
4. Assuming $d'$ is bias-free no matter what. It is criterion-independent only under the equal-variance model. If the zROC slope differs from 1, $d'$ from one $(H, F)$ pair depends on bias.
5. "Exponential and power are just two curve shapes." They make different claims about how forgetting works (constant vs. decreasing relative loss rate).
6. Judging a model only by its fit statistic. A small RMSD with a systematic residual pattern means the model's shape is wrong. Always plot the residuals.
7. Ignoring the noise model. A model is not just $f(x; \boldsymbol{\theta})$. The binomial noise of recall proportions is part of the model, and it tells you which differences are meaningful.
8. Changing the units of $t$ in the shifted power model and expecting the same shape.

---

### 1.1.9 Check your understanding

1. Name the five components of a model and give the SDT version of each. Answer Design (signal vs. noise trials; yes/no responses); parameters (free: $d'$, $c$; fixed: noise mean 0, both SDs 1); deterministic core ($H = \Phi(d'/2 - c)$, $F = \Phi(-d'/2 - c)$); noise model (hit and FA counts are binomial); linking assumption (respond "yes" when the evidence exceeds the criterion).

2. Why are the noise mean and the SDs in SDT fixed rather than free? Answer Because they are not identifiable: shifting both distributions and the criterion by the same amount, or stretching the whole axis, leaves every prediction unchanged. Fixing them (scaling constraints) defines the units in which $d'$ and $c$ are measured.

3. An observer has $H = 0.80$ and $F = 0.20$. Compute $d'$ and $c$. ($\Phi^{-1}(0.80) = 0.842$) Answer $d' = 0.842 - (-0.842) = 1.68$; $c = -\frac{1}{2}(0.842 - 0.842) = 0$, so the observer is unbiased.

4. What does equal-variance SDT predict about the zROC, and what does it mean if the empirical slope is 0.7? Answer A straight line with slope 1. A slope of 0.7 suggests the signal distribution is more variable than the noise distribution ($\sigma_s \approx 1/0.7 \approx 1.4$). Then single-point $d'$ depends on the criterion, and an unequal-variance model (or $d_a$) should be used.

5. What is the relative loss rate of each forgetting model, and why does it matter psychologically? Answer Exponential: $\lambda$, constant. Power: $d/(1+t)$, decreasing over time. The exponential says memories lose the same proportion of strength per day at every age; the power function says older memories become more resistant to further forgetting.

6. Which plot would you make to check quickly whether forgetting data look exponential? And for power? Answer Exponential: $\ln(\text{recall})$ against $t$ (semi-log), which should be a straight line. Power: $\ln(\text{recall})$ against $\ln(1+t)$ (log-log), which should be a straight line.

7. Give two reasons why fitting a straight line to log-transformed proportions is a poor estimation method. Answer (i) It breaks when an observed proportion is 0. (ii) It distorts the noise: small proportions become very variable on the log scale but are weighted equally, and the errors being minimized are on the wrong (log) scale. Fitting on the original scale with least squares or maximum likelihood avoids these problems.

---

### 1.1.10 Assignment

A1: From verbal to formal (1.0 point of 20; released Week 2, due at the start of the Week 4 lecture)

Scenario. Two research groups study forgetting of vocabulary learned in a foreign-language course. - Group C says: *"Memories decay at a constant rate."* - Group R says: *"Memories become more resistant to forgetting as they age."*

Both groups claim that the published data (recall drops quickly and then levels off) support their view.

Tasks. Submit a Jupyter notebook plus a PDF export.

1. Formalize (Group C and Group R). Write each verbal claim as a formal model with all five components from Lecture §1: design, free/fixed parameters with ranges, deterministic core, binomial noise model, and linking assumption. Justify in 2–3 sentences why each equation captures its verbal claim (use the relative loss rate).
2. Implement. Write Python functions that (a) return predicted recall probabilities and (b) simulate binomial recall counts for given intervals and number of items $n$.
3. Calibrate. Both groups agree that recall is 0.60 after 1 day and 0.35 after 7 days. Derive, by hand and in closed form, the parameters of each model that reproduce these two values exactly. Show the derivation (take logs) and verify it numerically.
4. Find where they diverge. With the calibrated parameters, plot both predicted curves from 0 to 90 days. For $n = 40$ items, compute at each interval a *discriminability index* $D(t) = \big(f_C(t) - f_R(t)\big)^2 / \text{Var}(t)$, where $\text{Var}(t) = \bar{f}(1-\bar{f})/n$ and $\bar{f}$ is the average of the two predictions. Plot $D(t)$.
5. Design an experiment. You can test each participant at only 4 retention intervals, chosen from {0.5, 1, 2, 4, 7, 14, 30, 60, 90} days. Choose the set that maximizes the summed $D(t)$, explain why it makes sense, and simulate one dataset from *each* model at your design. Show the two datasets on one plot.

6. Interpret (max. 300 words). Which data would favor Group C and which Group R? What does your analysis say about why both groups can claim support from "the published data"? Name one assumption of your models that could be wrong, and how it would change your conclusions.

Grading rubric (10 points, scaled to 1.0 of the final grade):

| Criterion | Points | Full marks require |
|---|---|---|
| Formalization (Task 1) | 2 | All five components for both models; correct parameter ranges; the link between equation and verbal claim is explained via the relative loss rate |
| Code (Task 2) | 2 | Correct, commented, reproducible (fixed seeds); functions are vectorized or clearly looped |
| Calibration (Task 3) | 2 | Correct closed-form derivation and numerical check for both models |
| Divergence and design (Tasks 4–5) | 2.5 | Correct $D(t)$; the chosen design is justified and actually maximizes the criterion; clear plots |
| Interpretation (Task 6) | 1.5 | Clear, specific, refers to your own results; names a real limitation |

Late submissions lose 10% per day (max. 3 days).

---

### 1.1.11 Further reading

- Farrell & Lewandowsky (2018), chapter 2. Building a model from a verbal theory; the components of a model and the role of parameters. *(Main reading.)*
- Lee & Wagenmakers (2014), chapter 11. Signal detection theory treated as a (Bayesian) cognitive model; we return to it in Weeks 9 and 13. Read the introductory part now, skip the code.
- Verguts (2022) and Anderson (2014): the chapters on model building and parameters [verify chapter numbers].

- *Background on forgetting curves:* Lee & Wagenmakers (2014), chapter 10 (memory retention) uses a closely related exponential retention model; we will fit it in a Bayesian way in Weeks 9–11.

*Preview for Week 3:* we drop linearization and fit models directly by minimizing squared error, which means learning how optimization algorithms search a parameter space.